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# Internal Phishing Simulation & SOC Investigation Lab

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*Full Build Guide – End-to-End Blue Team Project for SOC Portfolio*

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## 1. Introduction

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This document describes, in detail, the complete build of an internal phishing simulation and Security Operations Center (SOC) investigation lab. The goal of this project is to create a realistic, controlled environment where phishing emails can be simulated, delivered, and investigated using the same techniques and mindset that blue-team and SOC analysts use in real organizations.

The lab is built using

The lab is built using Windows Server, Windows 10, Kali Linux, hMailServer, Internet Information Services (IIS), Mozilla Thunderbird, and VMware Workstation. Together, these components form a small but realistic enterprise-like network where internal email flows, web traffic, firewall behaviour, and endpoint activity can be observed and analysed.

A phishing simulation, in this context, is the controlled creation and delivery of a deceptive email that mimics a real-world security alert or corporate communication. The purpose is not to harm, but to learn: to understand how such emails look, how they move through infrastructure, how users interact with them, and how defenders can detect and investigate them.

For SOC analysts, this type of lab is invaluable because it allows them to practise email header analysis, log correlation, incident reconstruction, and defensive configuration in a safe environment.

## 2. Lab Architecture and Design

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The lab is designed around a simple but powerful architecture. At the centre is a Windows Server virtual machine that plays two roles: it acts as an internal mail server using hMailServer, and it hosts a web server using IIS. This server represents the core infrastructure of a small organisation.

Alongside the server, there is a Windows 10 endpoint that represents a typical user workstation. This machine is used to receive and view emails using Thunderbird, and to simulate user behaviour such as opening messages and clicking links. A Kali Linux virtual machine is also included as an additional endpoint, used both as a client and as a system that can generate traffic and test network behaviour.

All virtual machines are created and managed in VMware Workstation and are connected to the same logical network using VMware NAT (VMnet8). This allows them to communicate internally while also having access to the internet through the host system. The internal domain used is microsoft.local, a private, non-routable domain chosen purely for lab purposes.

The choice of tools is deliberate:

- hMailServer – a lightweight but capable mail server suitable for internal labs.
- IIS – a standard Windows web server providing detailed logging.
- Thunderbird – a cross-platform email client that makes it easy to view full message headers and HTML content.
- Kali Linux – adds endpoint variety, reflecting mixed enterprise environments.
- VMware Workstation – provides a stable virtualisation platform for all systems.

## 3. Windows Server Setup

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### 3.1 Initial Server Configuration

The first step is to install Windows Server as a virtual machine in VMware Workstation. Once installed, the server is assigned a static IP address (e.g. 192.168.137.10) with a subnet mask of 255.255.255.0 and the VMware NAT gateway. A static IP is critical so that email clients and browsers can consistently reach the server. There is no requirement to set up a full Active Directory domain; the server remains in a workgroup.

### 3.2 Installing and Configuring hMailServer

After installation, the hMailServer Administrator is opened and a domain named microsoft.local is added. User accounts are created, including:

- admin@microsoft.local
- user2@microsoft.local
- account-security-noreply@microsoft.local

SMTP and IMAP are enabled on ports 25 and 143 respectively. Logging is enabled for SMTP, IMAP, and Application events. Log files are stored at:

```
C:\Program Files (x86)\hMailServer\Logs
```

### 3.3 Installing and Configuring IIS Web Server

IIS is installed via Server Manager by adding the Web Server (IIS) role. The default web root is:

```
C:\inetpub\wwwroot
```

A Microsoft login clone (sourced from the GitHub project by Octagon-Simon) is adapted and placed into this directory to serve as the phishing landing page. IIS logs are stored at:

```
C:\inetpub\logs\LogFiles\W3SVC1
```

These logs record each HTTP request including client IP, requested URL, timestamp, HTTP status code, and user agent string.

## 4. Networking Configuration (VMware)

### 4.1 VMware NAT Network

All VMs are attached to VMware's NAT network (VMnet8), which allows internet access through the host while maintaining a private internal subnet. Inspecting the host's VMware Network Adapter VMnet8 reveals the network range in use. For example:

- IPv4 Address: 192.168.137.1
- Subnet Mask: 255.255.255.0

This confirms a 192.168.137.0/24 subnet. The Windows Server is given the static IP 192.168.137.10; other VMs receive addresses via DHCP.

### 4.2 Connectivity Testing

Connectivity is verified with ping from each machine:

```
ping 192.168.137.10
```

Successful replies confirm that all machines can communicate. Failure indicates a network misconfiguration or firewall rule that must be corrected before proceeding.

## 5. Endpoint Configuration – Windows 10 and Kali

### 5.1 Installing Thunderbird on Windows 10

Mozilla Thunderbird is installed on the Windows 10 endpoint. An account is added using manual configuration with the following settings:

- Incoming – IMAP, server: 192.168.137.10, port: 143, no SSL, normal password

- Outgoing – SMTP, server: 192.168.137.10, port: 25, no SSL, normal password

After confirming connectivity, a test email is sent between internal accounts to verify correct operation.

## 5.2 Installing Thunderbird on Kali Linux

On Kali Linux, the package list is first updated and then Thunderbird ESR is installed:

```
sudo apt update
sudo apt install thunderbird-esr -y
```

If repository issues arise, the file `/etc/apt/sources.list` is updated to include:

```
deb http://http.kali.org/kali kali-rolling main contrib non-free non-free-
firmware
```

Thunderbird is then configured identically to the Windows 10 client. A round-trip test email confirms full integration.

## 6. PowerShell and Server-Side Commands

PowerShell is used throughout the lab on Windows machines to verify connectivity and inspect firewall rules. To test connectivity:

```
Test-Connection 192.168.137.10
```

To list all firewall rules:

```
Get-NetFirewallRule
```

These commands provide scripting capability and a deeper understanding of Windows network and security management.

## 7. Windows Firewall Configuration

### 7.1 Blocking ICMP (Ping) from Kali

To simulate defensive controls, Windows Firewall is configured to block ICMP echo requests. In Windows Defender Firewall with Advanced Security, the inbound rule File and Printer Sharing (Echo Request – ICMPv4-In) is disabled. Alternatively, a custom rule can be created targeting ICMPv4 Echo Request with an action of Block.

### 7.2 Verifying Firewall Behaviour

Before applying the rule, ping from Kali receives successful replies. After the rule is applied, ping results in Request timed out or Destination Host Unreachable. This demonstrates how firewall rules control network visibility and how configuration changes are reflected in observable behaviour.

## 8. Phishing Email Creation

### 8.1 Designing the HTML Email

The email mimics a Microsoft account security notification, containing a coloured header, body text describing unusual sign-in activity, and a prominent Review Activity button. The button links to the internal IIS server:

```
http://192.168.137.10/
```

The landing page uses HTML and CSS from the Microsoft login clone repository, providing a highly realistic sign-in interface.

## 8.2 CSS Used for the Login Clone

The CSS placed in the wwwroot directory defines fonts (Segoe UI and fallbacks), the background colour (#e2e5d3), the white auth wrapper with box shadow, responsive layout, and error variables — all matching the real Microsoft sign-in experience as closely as possible for lab purposes.

## 8.3 Sending the Email via hMailServer

In Thunderbird, the HTML email is composed using the sender address:

```
account-security-noreply@microsoft.local
```

The recipient is set to user2@microsoft.local and the subject to Microsoft account security alert. Thunderbird delivers the message via SMTP to hMailServer, which routes it to the recipient's inbox. The recipient sees the fully rendered HTML including the styled button.

# 9. SOC Investigation – Email Analysis

## 9.1 Analysing the Email in Thunderbird

The first investigative step is to view the full message source in Thunderbird. Key headers include:

- From – shows the display name and sender address (e.g. Microsoft <account-security-noreply@microsoft.local>).
- Return-Path – the actual return address; discrepancies may indicate spoofing.
- Received – shows the internal mail path; no external hops confirm the message is entirely internal.
- Message-ID – uniquely identifies the email and often includes the sending server's hostname, enabling log correlation.
- Authentication-Results – would show SPF, DKIM, and DMARC outcomes in a production environment.

## 9.2 Analysing hMailServer Logs

The hMailServer logs at C:\Program Files (x86)\hMailServer\Logs record SMTP connections, authentication, and delivery events. By matching the Message-ID and timestamp in the email headers against the log entries, the analyst can confirm the origin, verify that an authenticated internal account sent the message, and identify any potential account compromise.

# 10. SOC Investigation – Web and Network Evidence

## 10.1 IIS Web Logs

When the recipient clicks the Review Activity button, a browser launches and makes an HTTP GET request to the IIS server. The log file at C:\inetpub\logs\LogFiles\W3SVC1 records the client IP, URL, timestamp, HTTP status code (e.g. 200), and user agent. This entry confirms which endpoint clicked the link and precisely when.

## 10.2 Firewall and Network Behaviour

Firewall logs, where enabled, complement the web and email evidence by recording dropped packets and connection attempts. In this lab, ICMP blocking from Kali provides a tangible demonstration of how defensive controls limit network visibility and how such changes are directly observable in network behaviour.

## 11. Correlating Events Like a SOC Analyst

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A key SOC skill is correlating events across multiple systems to build a coherent incident timeline. In this lab, the following sequence can be reconstructed:

- 10:02 – Phishing email sent from account-security-noreply@microsoft.local; recorded in hMailServer SMTP log.
- 10:03 – Email delivered to user2@microsoft.local; confirmed by hMailServer delivery log.
- 10:05 – Recipient opens the email in Thunderbird; timing inferred from subsequent events.
- 10:06 – Recipient clicks Review Activity; IIS log records HTTP request from user's IP.
- 10:07 – Any additional network activity observed or blocked by the firewall.

Aligning timestamps and identifiers across email headers, hMailServer logs, IIS logs, and firewall records mirrors real SOC workflows where analysts piece together multi-source evidence to determine what happened, who was affected, and the scope of impact.

## 12. Skills Demonstrated

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This project demonstrates the following competencies:

- Infrastructure: Windows Server installation, static IP assignment, IIS role management.
- Mail Server Administration: hMailServer setup, domain and account configuration, protocol management, logging.
- Endpoint Configuration: Thunderbird IMAP/SMTP manual setup on Windows 10 and Kali Linux.
- Networking: VMware NAT subnet design, VM connectivity verification with ping and Test-Connection.
- Firewall Management: Locating and modifying Windows Firewall ICMPv4 rules; interpreting rule effects.
- Phishing Simulation: HTML email design, internal domain spoofing, login-clone landing page deployment.
- SOC Investigation: Email header analysis, hMailServer log inspection, IIS web log analysis, multi-source event correlation, and incident timeline reconstruction.

## 13. Conclusion

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This lab represents a complete internal phishing simulation and SOC investigation environment. It combines server configuration, endpoint setup, networking, firewall management, phishing email creation, and multi-source log analysis into a single coherent project.

The environment can be extended further by integrating a Security Information and Event Management (SIEM) platform such as Splunk, Wazuh, or Elastic. Logs from hMailServer, IIS, Windows, and network devices could be forwarded to the SIEM, where correlation rules and alerts

could be created to automatically detect phishing emails, suspicious clicks, or unusual network behaviour.

As it stands, this project is suitable for inclusion in a cybersecurity portfolio. It demonstrates practical, hands-on experience with tools and techniques directly relevant to SOC and blue-team roles, and reflects the mindset of a defender: analysing evidence, correlating events, and understanding how attacks and defences interact within a real network.